1. (a) $z_1^{(1)} = \theta_1^T x^{(1)} = 1\times5 + 2\times4 + 3\times(-2) = 7$
$z_2^{(1)} = \theta_2^T x^{(1)} = 1\times(-3) + 2\times2 + 3\times3 = 10$
$z_3^{(1)} = \theta_3^T x^{(1)} = 1\times4 + 2\times(-2) + 3\times3 = 9$
$z_1^{(2)} = \theta_1^T x^{(2)} = 1\times5 + (-1)\times4 + (-1)\times(-2) = 3$
$z_2^{(2)} = \theta_2^T x^{(2)} = 1\times(-3) + (-1)\times2 + (-1)\times3 = -8$
$z_3^{(2)} = \theta_3^T x^{(2)} = 1\times4 + (-1)\times(-2) + (-1)\times3 = 3$

$\Rightarrow p_2^{(1)} = \frac{e^{z_2^{(1)}}}{\sum_{j=1}^{3} e^{z_j^{(1)}}} = \frac{e^{10}}{e^7+e^{10}+e^9}$ $\quad p_3^{(2)} = \frac{e^{z_3^{(2)}}}{\sum_{j=1}^{3} e^{z_j^{(2)}}} = \frac{e^3}{e^3+e^{-8}+e^3}$

$\Rightarrow L(\theta) = -\frac{1}{2}(\log p_2^{(1)} + \log p_3^{(2)}) = -\frac{1}{2}[10 - \log(e^7+e^{10}+e^9) + 3 - \log(e^3+e^{-8}+e^3)] \approx 0.521$

(b) $p_1^{(1)} = e^7/(e^{10}+e^7+e^9) \approx 0.035$ $\quad p_1^{(2)} = e^3/(e^3+e^{-8}+e^3) \approx 0.5$
$p_2^{(1)} = e^{10}/(e^{10}+e^7+e^9) \approx 0.705$ $\quad p_2^{(2)} = e^{-8}/(e^3+e^{-8}+e^3) \approx 0$
$p_3^{(1)} = e^9/(e^{10}+e^7+e^9) \approx 0.259$ $\quad p_3^{(2)} = e^3/(e^3+e^{-8}+e^3) \approx 0.5$

$\nabla_{\theta_1} L^{(1)} = (p_1^{(1)} - 1(y^{(1)}=1))\, x^{(1)} = [0.035, 0.070, 0.105]^T$
$\nabla_{\theta_2} L^{(1)} = (p_2^{(1)} - 1(y^{(1)}=2))\, x^{(1)} = [-0.295, -0.589, -0.884]^T$
$\nabla_{\theta_3} L^{(1)} = (p_3^{(1)} - 1(y^{(1)}=3))\, x^{(1)} = [0.259, 0.519, 0.778]^T$
$\nabla_{\theta_1} L^{(2)} = (p_1^{(2)} - 1(y^{(2)}=1))\, x^{(2)} = [0.5, -0.5, -0.5]^T$
$\nabla_{\theta_2} L^{(2)} = (p_2^{(2)} - 1(y^{(2)}=2))\, x^{(2)} = [0, 0, 0]^T$
$\nabla_{\theta_3} L^{(2)} = (p_3^{(2)} - 1(y^{(2)}=3))\, x^{(2)} = [-0.5, 0.5, 0.5]^T$

$\Rightarrow -\nabla_{\theta_1} L = -\frac{1}{2}(\nabla_{\theta_1} L^{(1)} + \nabla_{\theta_1} L^{(2)}) = [-0.268, 0.215, 0.197]^T$
$-\nabla_{\theta_2} L = -\frac{1}{2}(\nabla_{\theta_2} L^{(1)} + \nabla_{\theta_2} L^{(2)}) = [0.147, 0.295, 0.442]^T$
$-\nabla_{\theta_3} L = -\frac{1}{2}(\nabla_{\theta_3} L^{(1)} + \nabla_{\theta_3} L^{(2)}) = [0.120, -0.509, -0.639]^T$

(c) $\theta_1' = \theta_1 + \alpha(-\nabla_{\theta_1} L) = [5, 4, -2]^T + 0.1\,[-0.268, 0.215, 0.197]^T = [4.973, 4.021, -1.980]$
$\theta_2' = \theta_2 + \alpha(-\nabla_{\theta_2} L) = [-3, 2, 3]^T + 0.1\,[0.147, 0.295, 0.442]^T = [-2.985, 2.029, 3.044]$
$\theta_3' = \theta_3 + \alpha(-\nabla_{\theta_3} L) = [4, -2, 3]^T + 0.1\,[0.120, -0.509, -0.639]^T = [4.012, -2.051, 2.936]$

2. (a) Let $z = \theta^T x$

$\Rightarrow \frac{\partial MSE}{\partial \theta_j} = \frac{\partial MSE}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial \theta_j}$

$\Rightarrow \frac{\partial MSE}{\partial h} = \frac{1}{2}\times 2\times(h(x)-y) = h(x)-y$

$\frac{\partial h}{\partial z} = \frac{-1(-e^{-z})}{(1+e^{-z})^2} = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{1+e^{-z}-1}{1+e^{-z}} = \frac{1}{1+e^{-z}}\left(1 - \frac{1}{1+e^{-z}}\right) = g(z)(1-g(z)) = h(x)(1-h(x))$

$\frac{\partial z}{\partial \theta_j} = x_j$

$\Rightarrow \frac{\partial MSE}{\partial \theta_j} = (h(x)-y)\,h(x)(1-h(x))\,x_j$

(b) ① Gradient Vanishing $\Rightarrow$ if $h(x)\to 0$ or $h(x)\to 1$ then $h(x)(1-h(x)) \to 0$ then $\nabla_\theta L \to 0$
$\Rightarrow \Delta\theta = \theta^{new} - \theta^{old} \to 0$
$\Rightarrow$ slow parameter updates
$\Rightarrow$ spend a lot of time training

② Non-Convexity $\Rightarrow$ let $x=1, y=1$ then $z=\theta$
$\Rightarrow L(\theta) = \frac{1}{2}(g(\theta)-1)^2$
$\Rightarrow L'(\theta) = (g(\theta)-1)(1-g(\theta))\,g(\theta)$
$\Rightarrow L''(\theta) = g(\theta)(1-g(\theta))^2(3g(\theta)-1)$
$\Rightarrow$ let $g(\theta) < \frac{1}{3}$ then $3g(\theta)-1 < 0$ then $L''(\theta) < 0$
$\Rightarrow$ Non-Convexity will involve local minima and saddle points
$\Rightarrow$ train result may not be global minimum

3. (a) let $\begin{cases} 0 < b_1 < 1 < r < 2 < b_2 < 3 \\ \phi(x) = (x-1.5)^2 \end{cases}$

$\Rightarrow \begin{cases} \phi(b_1) > 0.5^2 \\ \phi(b_2) > 0.5^2 \\ \phi(r) < 0.5^2 \end{cases}$

$\Rightarrow$ 0.5 can seperate b and r

(b) let $\begin{cases} b(x_b, y_b),\ |x_b| > 1,\ |y_b| > 1 \\ r(x_r, y_r),\ |x_r| < 1,\ |y_r| < 1 \\ \phi(x, y) = x^2 + y^2 \end{cases}$

$\Rightarrow \begin{aligned} \phi(b) &= x_b^2 + y_b^2 > 2 \\ \phi(r) &= x_r^2 + y_r^2 < 2 \end{aligned}$

$\Rightarrow$ 2 can seperate b and r

(c) let $\begin{cases} b(x_b, y_b),\ 1 < |x_b|, |y_b| < 2 \\ r_1(x_{r_1}, y_{r_1}),\ 0 < |x_{r_1}|, |y_{r_1}| < 1 \\ r_2(x_{r_2}, y_{r_2}),\ |x_{r_2}|, |y_{r_2}| > 2 \\ \phi(x, y) = (x-1.5)^2 + (y-1.5)^2 \end{cases}$

$\Rightarrow \begin{aligned} \phi(b) &= (x_{b_1}-1.5)^2 + (y_{b_1}-1.5)^2 < 0.5 \\ \phi(r_1) &= (x_{r_1}-1.5)^2 + (y_{r_1}-1.5)^2 > 0.5 \\ \phi(r_2) &= (x_{r_2}-1.5)^2 + (y_{r_2}-1.5)^2 > 0.5 \end{aligned}$

$\Rightarrow$ 0.5 can seperate b and r